THE ORDERED BAR COMPLEX AND E_1 PRIMACY

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Abstract. The five theorems of modular Koszul duality (bar-cobar adjunction, inversion, complementarity, genus universality, chiral Hochschild concentration) are proved in the literature for the symmetric bar complex $\overline{B}^{\Sigma}(\mathcal{A})$. The symmetric bar is a shadow: the Σ_n-coinvariant projection of the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, whose deconcatenation coproduct carries the full E_1-chiral coalgebra structure. This paper establishes E_1 primacy: the ordered bar is the primitive object; the symmetric bar is derived by averaging; and the five theorems are the invariants that survive the quotient.

The averaging map $\mathrm{av}\colon \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is a surjective morphism of dg Lie algebras that sends the ordered Maurer–Cartan element $\Theta^{\mathsf{E}_1}_{\mathcal{A}}$ to the modular $\Theta_{\mathcal{A}}$, projects the classical r-matrix $r(z)$ to $\kappa(\mathcal{A})$, and projects the Drinfeld–Kontsevich associator Φ_{KZ} to the cubic shadow $\mathfrak{C}(\mathcal{A})$. The map does not split as dg Lie algebras: the kernel at degree 3 contains the linearised associator $[\Omega_{12}, \Omega_{23}]$, and the space of Lie sections compatible with the Maurer–Cartan equation is a torsor for the Grothendieck–Teichmüller group GRT_1. At genus 1, a fourth obstruction to splitting arises from the quasi-modular anomaly $E_2 \mapsto \widehat{E}_2$.

For the principal examples: Heisenberg $\mathcal{H}_k$ has $r(z) = k/z$ and $\kappa(\mathcal{H}_k) = k$; affine Kac–Moody $\widehat{\mathfrak{g}}_k$ has $r(z) = k\Omega_{\mathfrak{g}}/z$ (trace-form convention; vanishing at $k = 0$) and $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$, the Sugawara shift $\dim(\mathfrak{g})/2$ arising from the simple-pole self-contraction; Virasoro Vir_c has $r(z) = (c/2)/z^3 + 2T/z$ and $\kappa(\mathrm{Vir}_c) = c/2$. The kernel of av at degree 3 contains the KZ associator; no symmetric invariant recovers this datum. We prove the E_1 primacy theorem, state the five ordered counterparts $\mathrm{A}^{\mathsf{E}_1}$–$\mathrm{H}^{\mathsf{E}_1}$, and exhibit the formality bridge and its failure.

1. Introduction: the averaging map and what it loses

The bar complex attached to an augmented associative algebra admits two very different presentations. In the first, one takes the reduced tensor coalgebra $T^c(s^{-1}\bar{A})$ with its deconcatenation coproduct; the bar differential is built from the associative product in consecutive slots. In the second, one passes to Σ_n-coinvariants and works with the symmetric bar complex, which is the natural home for commutative or E_∞-algebras and for the Francis–Gaitsgory chiral bar. The historical order in chiral algebra and factorization homology has been: symmetric bar first, ordered refinement second. This paper reverses the order of presentation, and with it the status of the two objects.

1.1. The thesis. The thesis is that the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the primitive object of modular Koszul duality, and that the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ is a derived shadow obtained by averaging. The averaging map is a surjective morphism of dg Lie algebras; all five foundational theorems of modular Koszul duality (Theorems A through D, together with the chiral Hochschild theorem H) lift to the ordered side, and each symmetric theorem is the Σ_n-coinvariant image of its ordered counterpart. arithmetic, and physical: the spectral r-matrix, the KZ connection, the Drinfeld associator, and the full braiding of tensor categories of modules all live on the ordered side.

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Three principles motivate the inversion. The first is operadic: Stasheff's associahedra K_n [20] carry more information than the symmetric groups Σ_n, and the Getzler–Jones [9] theory of E_n-operads makes E_1 the lowest rung of a ladder whose every step is non-commutative. The second is representation-theoretic: the classical r-matrix of an affine Kac–Moody algebra at level k is the content of the degree-2 genus-0 Maurer–Cartan element on the ordered side, and its Σ_2-average recovers the degree-2 scalar shadow: for Heisenberg this is κ, while for non-abelian affine Kac–Moody it is $\kappa_{\mathrm{dp}} = k\dim(\mathfrak{g})/(2h^\vee)$, with the full κ obtained after adding the Sugawara shift $\dim(\mathfrak{g})/2$. The third is physical: the line-operator sector of a chiral conformal theory is naturally open and ordered, while the closed (bulk) sector is symmetric; the averaging map is the operadic image of the open-to-closed functor.

We work throughout with the conventions of the monograph [14], which we refer to as Volume I. Everything is stated with OPE modes rather than lambda-brackets; conformal weights are denoted h, and desuspended degrees $|s^{-1}v| = |v| - 1$ (the bar complex lowers degree). The grading is cohomological.

1.2. **Notation and conventions.** Let k be a field of characteristic zero, X a smooth algebraic curve over k, and $\mathcal{A}$ an augmented cyclic chiral algebra on X in the sense of Beilinson–Drinfeld [2]. We write $\bar{\mathcal{A}}$ for the augmentation ideal $\ker(\varepsilon\colon \mathcal{A} \to k)$. All bar constructions are defined on $\bar{\mathcal{A}}$, not on $\mathcal{A}$ itself [14]. Symmetric groups Σ_n act on tensor powers in the standard way; coinvariants are left exact and are denoted with a lower subscript, $(-)_{\Sigma_n}$. The ordered and symmetric Fulton–MacPherson compactifications are written $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ and $\overline{\mathrm{FM}}_n(\mathbb{C})$ respectively; the former is the Ran-space blow-up along ordered collision diagonals.

1.3. **Summary of results.** The main results are:

(i) The E_1 *convolution dg Lie algebra* $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$ (Definition 3.4) and its ordered Maurer–Cartan element $\Theta^{\mathsf{E}_1}_{\mathcal{A}}$ (Theorem 3.5).

(ii) The *averaging map* $\mathrm{av}\colon \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ as a surjective dg Lie morphism (Theorem 4.2), with the projection formula $\mathrm{av}(\Theta^{\mathsf{E}_1}_{\mathcal{A}}) = \Theta_{\mathcal{A}}$ (Theorem 4.6).

(iii) The E_1 *primacy theorem* (Theorem 5.1): surjectivity, MC projection, bracket preservation, and non-splitting of av.

(iv) The *non-splitting proposition* (Proposition 4.7): the kernel of av is controlled by GRT_1 at genus 0 and by the quasi-modular anomaly at genus 1.

(v) The *five ordered theorems* $\mathrm{A}^{\mathsf{E}_1}$–$\mathrm{H}^{\mathsf{E}_1}$ (Theorems 5.4–5.8), whose av-images are the classical five theorems.

(vi) Complete degree-2 computations for Heisenberg, affine Kac–Moody, and Virasoro (Section 8).

2. The ordered bar complex

2.1. **Three coalgebra structures on $T(V)$.** Before defining the ordered bar complex we distinguish three cooperadic structures carried by a sum of tensor powers. All three arise somewhere in modular Koszul duality; only one equips the ordered bar.

Definition 2.1 (The three coalgebra structures)**.** Let V be a graded vector space and $T(V) = \bigoplus_{n\geq 0} V^{\otimes n}$ its tensor space. Three cooperad structures arise naturally:

(i) The *Harrison coLie* cooperad Lie^c, whose degree-n component has dimension $(n-1)!$ and whose coproduct decomposes a Lie word into pairs of sub-Lie-words using the Harrison differential.

(ii) The *coshuffle cooperad* Sym^c, whose degree-n coproduct produces 2^n summands by the shuffle formula

$$\Delta_{\mathrm{sh}}(v_1\cdots v_n) = \sum_{I\sqcup J=\{1,\dots,n\}} \varepsilon(I,J)\,(v_I)\otimes(v_J),$$

with the sum running over all 2^n bipartitions and $\varepsilon(I,J)$ the Koszul sign.

(iii) The *deconcatenation cooperad* T^c, whose degree-n coproduct produces $n+1$ summands by splitting the word at a single position:

$$\Delta(v_1\cdots v_n) = \sum_{i=0}^{n}(v_1\cdots v_i)\otimes(v_{i+1}\cdots v_n).$$

The three structures are *not* interconvertible without loss of information. At degree 3 the Harrison differential sees 2 terms, the coshuffle sees 8, and deconcatenation sees 4. Only the last is ordered and strictly associative at the coalgebra level.

Proposition 2.2 (Coshuffle is not deconcatenation)**.** *Let V be nonzero. The coshuffle coproduct on* $\mathrm{Sym}(V)$ *and the deconcatenation coproduct on $T^c(V)$ are distinct cooperadic structures: they have different degree-n term counts (2^n vs. $n+1$), different symmetry properties (the coshuffle is cocommutative; deconcatenation is not), and different primitive subspaces.*

Proof. Cocommutativity of the coshuffle follows from the involution $I\leftrightarrow J$ on bipartitions. Deconcatenation does not admit such an involution: swapping $(v_1\cdots v_i)$ with $(v_{i+1}\cdots v_n)$ does not preserve the ordered word. The term counts are elementary. □

Remark 2.3 (Where each structure appears). The Harrison coLie cooperad Lie^c underlies the chiral Lie bar complex used by Francis–Gaitsgory [8], where only the zeroth product $a_{(0)}b$ of the OPE is visible. The coshuffle cooperad Sym^c underlies the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ on the Ran space, where factorization is implemented by the topological coproduct on unordered configuration space. The deconcatenation cooperad T^c underlies the ordered bar $\overline{B}^{\mathrm{ord}}(\mathcal{A})$, where the coproduct splits a word along a single internal edge and remembers the ordering on the collision points. The averaging map av of Section 4 descends from T^c to Sym^c by external R-twisted Σ_n-coinvariants $\overline{B}^{\Sigma}_n(\mathcal{A})\simeq(\overline{B}^{\mathrm{ord}}_n(\mathcal{A}))^{R\text{-}\Sigma_n\text{-coinv}}$ (the R-twist is trivial on the pole-free subclass, where naive Σ_n-coinvariants suffice;); it does not factor through Harrison unless one further restricts to the Eulerian weight-1 summand.

2.2. The ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})$.

Definition 2.4 (The ordered bar complex)**.** Let $\mathcal{A}$ be a strongly admissible augmented E_1-chiral algebra on X, with augmentation ideal $\bar{\mathcal{A}}=\ker\varepsilon$. The *ordered bar complex* is the graded vector space

$$\overline{B}^{\mathrm{ord}}(\mathcal{A}) \;:=\; T^c(s^{-1}\bar{\mathcal{A}}) \;=\; \bigoplus_{n\ge 0}(s^{-1}\bar{\mathcal{A}})^{\otimes n},$$

equipped with:

(a) The *deconcatenation coproduct*

$$\Delta\big([a_1|\cdots|a_n]\big) \;=\; \sum_{i=0}^{n}[a_1|\cdots|a_i]\otimes[a_{i+1}|\cdots|a_n],$$

making $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ a cofree conilpotent coalgebra over T^c.

(b) The *bar differential* d_{bar}, obtained by extracting OPE collision residues at consecutive pairs of points in the ordered configuration space $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$:

$$d_{\mathrm{bar}}([a_1|\cdots|a_n]) \;=\; \sum_{i=1}^{n-1} \pm\,[a_1|\cdots|a_i \star a_{i+1}|\cdots|a_n],$$

where $a_i \star a_{i+1}$ denotes the collision residue (the coefficient of the simple pole in the OPE of $a_i(z)$ and $a_{i+1}(w)$ against the Arnold form $\eta = d\log(z-w)$).

The desuspension s^{-1} lowers the degree by one: $|s^{-1}a| = |a| - 1$. The augmentation ideal is essential; using $\mathcal{A}$ in place of $\bar{\mathcal{A}}$ would include the unit and break the construction.

Theorem 2.5 (Ordered bar differential squared)**.** *The ordered bar differential satisfies $d_{\mathrm{bar}}^2 = 0$.*

Proof. The argument is a codimension-two face pairing on $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{C})$ with the Arnold relation supplying the required cancellation between the two ways of contracting an adjacent pair of collisions. Each codimension-2 stratum is reached by two codimension-1 paths (contract e_1 then e_2, or e_2 then e_1), and the edge-orientation determinant supplies a minus sign between them. The cancellation uses only the associativity of face composition in the face poset (Getzler–Kapranov [10]), which holds for ordered configurations by the same argument as for unordered stable graphs. □

2.3. **The classical r-matrix as degree-2 bar data.** The binary collision residue of the ordered bar differential extracts the classical r-matrix. This is the degree-2 component of the E_1 Maurer–Cartan element, and it is the central object of the E_1 theory.

Definition 2.6 (The classical r-matrix from the bar complex)**.** Let $\mathcal{A}$ be a cyclic chiral algebra on X with OPE $a(z)\,b(w) \sim \sum_{n\geq 0}(a_{(n)}b)(w)/(z-w)^{n+1}$. The *classical r-matrix* of $\mathcal{A}$ is the degree-2 genus-0 component of the E_1 Maurer–Cartan element:

$$r_{\mathcal{A}}(z) \;:=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta^{\mathsf{E}_1}_{\mathcal{A}}) \;\in\; \mathrm{End}_{\mathcal{A}}(2) \otimes \mathcal{O}(*\Delta).$$

Concretely, $r(z)$ is the meromorphic function of the spectral parameter $z = z_1 - z_2$ obtained by extracting the OPE against the Arnold form $\eta_{12} = d\log(z_1 - z_2)$ on the two-point ordered configuration space.

Remark 2.7 (Pole absorption). The r-matrix has poles one order lower than the OPE, because the logarithmic form $d\log(z)$ absorbs one pole: the OPE of two fields with a pole of order p produces an r-matrix with a pole of order $p-1$. This is the $d\log$ absorption rule: Heisenberg OPE $\sim 1/(z-w)^2$ gives $r \sim 1/z$; Virasoro OPE $\sim 1/(z-w)^4$ gives $r \sim 1/z^3$.

Example 2.8 (Heisenberg ordered bar)**.** Let $\mathcal{H}_k$ be the Heisenberg chiral algebra at level k, generated by a single field J of conformal weight $h = 1$. The desuspended generator has degree $|s^{-1}J| = 0$. At degree 2 the ordered bar component is

$$(s^{-1}\bar{\mathcal{H}}_k)^{\otimes 2} \;\cong\; k \cdot [J|J],$$

a one-dimensional space. The bar differential extracts the collision residue of $J(z_1)J(z_2) \sim k/(z_1 - z_2)^2$ against $\eta = d\log(z_1 - z_2)$, producing a single simple pole $k/(z_1 - z_2)$. The r-matrix is

$$r^{\mathrm{Heis}}(z) \;=\; k/z. \tag{2.1}$$

At $k = 0$ the r-matrix vanishes: the algebra becomes trivially commutative. At degree 3, the Arnold relation on $\overline{\mathrm{FM}}_3(\mathbb{C})$ gives $d_{\mathrm{bar}}^2 = 0$: the three collision divisors contribute $k \cdot (\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{13} + \eta_{13} \wedge \eta_{12}) = 0$.

Remark 2.9 (Heisenberg R-matrix convention: curved vs flat). The Heisenberg algebra $\mathcal{H}_k$ admits two standard presentations of the ordered bar R-datum: the *flat* (classical, linear) form $r(z) = k/z$ used throughout this note, and the *curved* (quantum, exponentiated) form $R(z) = \exp(k\;\;/z)$ used in the main chapter and in five companion standalones. They are related by the exponential map: $R(z) = \exp(\;\;r(z))$, so that the classical r-matrix is the ^{0} coefficient of $\log R$. At every order in the exponentiation satisfies the quantum Yang–Baxter equation because $r(z)$ is a Casimir (its two tensor slots commute); the curved form is the trivialization of the twist, the flat form its infinitesimal shadow. This is the curved $\leftrightarrow$ flat Heisenberg equivalence (Costello–Witten twist trivialization); at $= 0$ the classical limit recovers $r(z)$.

Example 2.10 (Affine Kac–Moody ordered bar). Let $\widehat{\mathfrak{g}}_k$ be the affine Kac–Moody algebra at level k for a simple Lie algebra $\mathfrak{g}$ of dimension $\dim(\mathfrak{g})$ and dual Coxeter number $h^\vee$. The OPE has both simple and double poles: $J^a(z)J^b(w) \sim k\kappa^{ab}/(z-w)^2 + f^{ab}{}_c J^c(w)/(z-w)$, where κ^{ab} is the Killing form and $f^{ab}{}_c$ the structure constants. The r-matrix in the trace-form convention is

$$r^{\mathrm{KM}}(z) \;=\; k\,\Omega_{\mathfrak{g}}/z, \tag{2.2}$$

where $\Omega_{\mathfrak{g}} = \sum_a t^a \otimes t_a \in \mathfrak{g} \otimes \mathfrak{g}$ is the inverse Killing form Casimir. The level prefix k is essential: at $k = 0$ the r-matrix vanishes (the algebra becomes abelian in the sense that the double pole, which governs the r-matrix, is zero). At the critical level $k = -h^\vee$, the Sugawara construction fails; the behaviour of the r-matrix at this locus is controlled by the Feigin–Frenkel centre.

The KZ normalization $r(z) = \Omega_{\mathfrak{g}}/((k + h^\vee)z)$ is equivalent at generic k via the bridge identity $k\,\Omega_{\mathrm{tr}} = \Omega/(k + h^\vee)$.

Example 2.11 (Virasoro ordered bar). Let Vir_c be the Virasoro algebra at central charge c. The OPE $T(z)T(w) \sim (c/2)/(z-w)^4 + 2T(w)/(z-w)^2 + \partial T(w)/(z-w)$ has a quartic pole. The $d\log$ absorption gives a cubic-plus-simple r-matrix:

$$r^{\mathrm{Vir}}(z) \;=\; (c/2)/z^3 + 2T/z.$$

This is cubic plus simple, not quartic.

3. The E_1 convolution algebra and its MC element

3.1. **The ribbon modular operad.** The combinatorial unit of the commutative modular operad $\mathcal{M}_{\mathrm{Com}}$ is a stable graph with symmetric vertex labels: ordering data is erased before composition starts. One modular operad retains the ordering without breaking the genus decomposition: the ribbon modular operad $\mathcal{M}_{\mathrm{Ass}}$.

Definition 3.1 (Ribbon modular operad [3, 10]). The *ribbon modular operad* $\mathcal{M}_{\mathrm{Ass}}$ has components

$$\mathcal{M}_{\mathrm{Ass}}(g,n) \;:=\; C_*\big(\overline{\mathcal{M}}^{\mathrm{rib}}_{g,n}\big), \qquad 2g - 2 + n > 0,$$

where $\overline{\mathcal{M}}^{\mathrm{rib}}_{g,n}$ classifies stable genus-g Riemann surfaces with n parametrised boundary circles. A ribbon structure on a stable graph is a cyclic ordering of half-edges at each vertex. In genus 0: $\mathcal{M}_{\mathrm{Ass}}(0,n) = C_*(K_n)$, where K_n is the Stasheff associahedron of dimension $n-2$, and grafting is face inclusion. The composition maps are self-sewing $\xi_{ij} : \mathcal{M}_{\mathrm{Ass}}(g,n) \to \mathcal{M}_{\mathrm{Ass}}(g+1, n-2)$ and grafting $\circ_i : \mathcal{M}_{\mathrm{Ass}}(g_1,n_1) \otimes \mathcal{M}_{\mathrm{Ass}}(g_2,n_2) \to \mathcal{M}_{\mathrm{Ass}}(g_1+g_2, n_1+n_2-2)$.

Definition 3.2 (Feynman transform $F\mathrm{Ass}$). The *associative Feynman transform* $F\mathrm{Ass}$ is the modular cooperad with (g,n)-component

$$(F\mathrm{Ass})(g,n) \;:=\; \bigoplus_{\Gamma \in \mathsf{Gr}^{\mathrm{rib}}_{g,n}} \frac{1}{|\operatorname{Aut}(\Gamma)|} \det\big(E(\Gamma)\big) \;\otimes\; \bigotimes_{v \in V(\Gamma)} \mathcal{M}_{\mathrm{Ass}}\big(g(v), n(v)\big)^\vee,$$

summed over connected ribbon graphs of genus g with n legs. The differential $D_{F\mathrm{Ass}} = \sum_{e\in E(\Gamma)} \pm \xi_e^*$ is dual to edge contraction. This differential satisfies $D^2_{F\mathrm{Ass}} = 0$ by the standard codimension-2 cancellation argument [10].

Theorem 3.3 (Coinvariant projection [10, Thm. 4.9])**.** *The composite of the ribbon-forgetting and external Σ_n-coinvariant projections*

$$(F\mathrm{Ass})(g,n) \xrightarrow{\pi_{\mathrm{rib}}} (F\mathrm{Ass})(g,n)/\sim_{\mathrm{rib}} \xrightarrow{\pi_{\Sigma_n}} (F\mathrm{Com})(g,n)$$

is a quasi-isomorphism of modular cooperads.

3.2. The E_1 convolution dg Lie algebra.

Definition 3.4 (The E_1 modular convolution algebra)**.** Let $\mathcal{A}$ be a cyclic E_1-chiral algebra. The E_1 *modular convolution dg Lie algebra* is

$$\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} := \prod_{\substack{g,n \\ 2g-2+n>0}} \mathrm{Hom}\big(\mathcal{M}_{\mathrm{Ass}}(g,n),\ \mathrm{End}_{\mathcal{A}}(n)\big).$$

The Hom carries *no* Σ_n-equivariance. This is the structural distinction from the modular convolution algebra

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \prod_{\substack{g,n \\ 2g-2+n>0}} \mathrm{Hom}_{\Sigma_n}\big(\mathcal{M}_{\mathrm{Com}}(g,n),\ \mathrm{End}_{\mathcal{A}}(n)\big),$$

whose Hom-source uses the symmetric cooperad Sym^c obtained from T^c by external Σ_n-coinvariants. The dg Lie structure on $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$ is inherited from $F\mathrm{Ass}$: the differential D comes from $D_{F\mathrm{Ass}}$ (edge contraction), and the bracket $[-,-]$ comes from ribbon-graph composition.

Theorem 3.5 (E_1 Maurer–Cartan element)**.** *The element*

$$\Theta^{\mathsf{E}_1}_{\mathcal{A}} := D^{\mathsf{E}_1}_{\mathcal{A}} - d_0 \in \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}},$$

where $D^{\mathsf{E}_1}_{\mathcal{A}}$ is the ordered genus-completed bar differential and d_0 the background differential, satisfies the Maurer–Cartan equation

$$D\,\Theta^{\mathsf{E}_1}_{\mathcal{A}} + \tfrac{1}{2}[\Theta^{\mathsf{E}_1}_{\mathcal{A}}, \Theta^{\mathsf{E}_1}_{\mathcal{A}}] = 0 \tag{3.1}$$

in the ordered dg Lie algebra $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$.

Proof. The identity $(D^{\mathsf{E}_1}_{\mathcal{A}})^2 = 0$ is equivalent to the MC equation (3.1) upon writing $D^{\mathsf{E}_1}_{\mathcal{A}} = d_0 + \Theta^{\mathsf{E}_1}_{\mathcal{A}}$. The square-zero property is the ordered bar-intrinsic construction: $D^{\mathsf{E}_1}_{\mathcal{A}}$ is built from the associative Feynman transform, and $D^2_{F\mathrm{Ass}} = 0$ forces the induced coderivation to square to zero. □

Proposition 3.6 (E_1 shadow at each degree)**.** *The degree-r component of $\Theta^{\mathsf{E}_1}_{\mathcal{A}}$ at genus* 0 *is an element $r_r \in \mathrm{End}(V^{\otimes r}) \otimes H^{r-2}(K_r) \otimes \mathcal{O}(*\Delta)$, whose Σ_r-coinvariant is the E_∞ shadow at degree r:*

Degree	E_1 shadow	E_∞ shadow
$r=2$	$r(z)$ *(classical r-matrix)*	$\kappa(\mathcal{A})$
$r=3$	$r_3(z_1,z_2)$ *(associator shadow)*	$\mathfrak{C}(\mathcal{A})$
$r=4$	$r_4(z_1,z_2,z_3)$ *(quartic shadow)*	$\mathfrak{Q}(\mathcal{A})$

The MC equation at degree r is the ∂K_{r+1}-identity: classical Yang–Baxter at $r=2$ (two faces of K_3), pentagon at $r=3$ (five faces of K_4), and the quartic R-matrix identity at $r=4$.

Proof. At genus 0, $\mathcal{M}_{\mathrm{Ass}}(0,r) = C_*(K_r)$. The MC equation projected to degree r is the equation on the boundary of K_{r+1}, whose faces are precisely the binary trees contributing to the r-point collision structure. At $r = 2$, K_3 is an interval with two vertices (the two binary trees on three leaves), and the boundary identity is the classical Yang–Baxter equation. □

4. The averaging map: surjectivity, MC projection, non-splitting

4.1. Definition of av.

Definition 4.1 (The averaging map). With the notation of Definition 3.4, the *averaging map* is the dg Lie morphism

$$\mathrm{av}\colon \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, \qquad \mathrm{av}(\phi)(g,n) \;:=\; \frac{1}{n!}\sum_{\sigma\in\Sigma_n} \sigma\cdot\big(\phi(g,n)\circ\iota^{\mathrm{rib}}_{g,n}\big),$$

where $\iota^{\mathrm{rib}}_{g,n}\colon \mathcal{M}_{\mathrm{Com}}(g,n)\to\mathcal{M}_{\mathrm{Ass}}(g,n)$ is any section of the ribbon-forgetting quotient (Theorem 3.3). The result is Σ_n-invariant by construction and independent of the choice of section up to boundary terms. This is the Reynolds operator for the external Σ_n-action: it takes external coinvariants after pullback along ι^{rib}.

Theorem 4.2 (av is a surjective dg Lie morphism). *The averaging map* $\mathrm{av}\colon \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ *is a surjective morphism of dg Lie algebras.*

Proof. Surjectivity: every Σ_n-invariant homomorphism from $\mathcal{M}_{\mathrm{Com}}(g,n)$ lifts, via any section of the ribbon-forgetting quotient, to a homomorphism from $\mathcal{M}_{\mathrm{Ass}}(g,n)$. Compatibility with the differential: the ribbon edge-contraction differential descends to the symmetric edge-contraction differential under *R-twisted* external Σ_n-coinvariants $(-)^{R\text{-}\Sigma_n\text{-coinv}}$ –- the twist is trivial on the pole-free subclass ($R=\mathrm{id}$), reducing to naive coinvariants, and carries the Heisenberg monodromy $R(z)=\exp(k\ \ /z)$ on curved families. Under this twist the comparison between ribbon and commutative Feynman transforms [10] gives the differential compatibility. Compatibility with the Lie bracket: symmetrisation commutes with the convolution bracket on cooperadic Homs because operad composition in $F\mathrm{Ass}$ is Σ_n-equivariant. □

4.2. Degree-2 averaging: the r-matrix projects to κ.

Theorem 4.3 (Degree-2 averaging). *Let* $r(z)\in\mathrm{End}(V\otimes V)\otimes\mathcal{O}(*\Delta)$ *be the genus-*0 *degree-*2 E_1 *shadow of $\mathcal{A}$ (the classical r-matrix in the pre-dualisation convention). Then:*

$$\mathrm{av}_{n=2}\big(r(z)\big) \;=\; \begin{cases} \kappa(\mathcal{A}), & \textit{for abelian Heisenberg families,}\\[4pt] \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) \;:=\; \dfrac{k\ \dim(\mathfrak{g})}{2h^\vee}, & \textit{for non-abelian affine KM}\\ & \textit{in the trace-form convention.}\end{cases}$$

In the affine Kac–Moody case the full modular characteristic is

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim(\mathfrak{g})}{2} \;=\; \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}, \tag{4.1}$$

where the extra $\dim(\mathfrak{g})/2$ *is the Sugawara simple-pole self-contraction through the adjoint Casimir eigenvalue* $2h^\vee$.

Proof. The degree-2 MC component is, by definition, an element $r(z)\in\mathrm{End}(V^{\otimes 2})\otimes\mathcal{O}(*\Delta)$ in the pre-dualisation convention. Averaging over Σ_2 projects onto the trivial isotypic component. For abelian Heisenberg families this scalar coincides with the full modular characteristic. For non-abelian affine Kac–Moody in the trace-form convention $r(z)=k\,\Omega_{\mathfrak{g}}/z$, the averaging sees only the double-pole channel and yields $\kappa_{\mathrm{dp}}=k\ \dim(\mathfrak{g})/(2h^\vee)$; the simple-pole self-contraction contributes the additional Sugawara term $\dim(\mathfrak{g})/2$, so $\kappa = \kappa_{\mathrm{dp}} + \dim(\mathfrak{g})/2$. The example computations are recorded in [14, Chapter 5]; each formula is verified by three independent paths: pole-order analysis, the limiting case at $k = 0$ for the trace-form r-matrix, and cross-family consistency against the landscape census. □

Remark 4.4 (Concrete examples). The principal families:

- Heisenberg $\mathcal{H}_k$: $r(z) = k/z$, $\kappa(\mathcal{H}_k) = k$.
- Affine Kac–Moody $\widehat{\mathfrak{g}}_k$: $r(z) = k\,\Omega_{\mathfrak{g}}/z$ (the level k survives $d\log$ absorption; at $k = 0$ the r-matrix vanishes identically), $\mathrm{av}(r(z)) = \kappa_{\mathrm{dp}} = k\ \dim(\mathfrak{g})/(2h^\vee)$, and $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$.
- Virasoro Vir_c: $r(z) = (c/2)/z^3 + 2T/z$ (cubic plus simple, not quartic), and $\kappa(\mathrm{Vir}_c) = c/2$.

4.3. Degree-3 averaging: the associator projects to the cubic shadow.

Theorem 4.5 (Degree-3 averaging). *Let $\Phi_{\mathrm{KZ}}(\mathcal{A}) \in \mathrm{End}(V^{\otimes 3})$ be the genus-0 degree-3 Maurer–Cartan component, which for affine Kac–Moody $\widehat{\mathfrak{g}}_k$ is the Drinfeld–Kontsevich KZ associator. Then*

$$\mathrm{av}_{n=3}\big(\Phi_{\mathrm{KZ}}(\mathcal{A})\big) \;=\; \mathfrak{C}(\mathcal{A}),$$

where $\mathfrak{C}(\mathcal{A})$ is the cubic shadow of Volume I, equivalently the Σ_3-coinvariant of the associator. The antisymmetric component $[\Omega_{12}, \Omega_{23}]$ lies entirely in $\ker(\mathrm{av})$: it is antisymmetric under the transposition $(1 \leftrightarrow 3)$, so its Σ_3-average vanishes. Only the symmetric products $\Omega_{ij}\Omega_{k\ell} + \Omega_{k\ell}\Omega_{ij}$ contribute to $\mathfrak{C}(\mathcal{A})$.

Proof. The Σ_3-isotypic decomposition of Φ_{KZ} in $\mathrm{End}(V^{\otimes 3})$ splits into the trivial, sign, and standard representations. The Reynolds operator $\mathrm{av} = (1/6)\sum_{\sigma\in\Sigma_3}\sigma$ projects onto the trivial isotypic component. A direct check on $[\Omega_{12}, \Omega_{23}]$ shows it is antisymmetric under $(1 \leftrightarrow 3)$ and therefore has trivial isotypic component zero. The identification of the coinvariant with the cubic shadow $\mathfrak{C}(\mathcal{A})$ is [14, Theorem D(iii)]. □

4.4. MC projection.

Theorem 4.6 (MC projection under av). *Under averaging:*

$$\mathrm{av}\big(\Theta^{\mathsf{E}_1}_{\mathcal{A}}\big) \;=\; \Theta_{\mathcal{A}}.$$

In particular, the shadow obstruction tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \ldots)$ is the degree-by-degree Σ_n-coinvariant image of the R-matrix tower $(r(z), r_3, r_4, \ldots)$.

Proof. $\mathrm{av}(D^{\mathsf{E}_1}_{\mathcal{A}}) = D_{\mathcal{A}}$ because the ordered genus-completed bar differential on $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ descends to $\overline{B}^{\Sigma}(\mathcal{A})$ under R-twisted Σ_n-coinvariants, and $\mathrm{av}(d_0) = d_0$. On the pole-free subclass the twist is trivial and descent reduces to naive coinvariants; for Heisenberg, affine Kac–Moody, and higher E_∞-chiral families with poles, the twist carries the monodromy $R(z)$ along elementary transpositions, and the descent remains well-defined by quantum Yang–Baxter. Therefore

$$\mathrm{av}(\Theta^{\mathsf{E}_1}_{\mathcal{A}}) = \mathrm{av}(D^{\mathsf{E}_1}_{\mathcal{A}} - d_0) = D_{\mathcal{A}} - d_0 = \Theta_{\mathcal{A}}.$$ □

4.5. Non-splitting.

Proposition 4.7 (Non-splitting of av). *The short exact sequence of dg Lie algebras*

$$0 \longrightarrow \ker(\mathrm{av}) \longrightarrow \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \xrightarrow{\mathrm{av}} \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \longrightarrow 0 \tag{4.2}$$

does not split as dg Lie algebras. Four independent obstructions prevent a section:

(i) Fixed-degree triviality. *At each fixed degree n, the inclusion section $s\colon \mathrm{End}^{\Sigma_n}(V^{\otimes n}) \hookrightarrow \mathrm{End}(V^{\otimes n})$ is a Lie subalgebra embedding, so the extension 2-cocycle vanishes. The extension is Lie-trivial at each degree separately.*

(ii) Cross-degree obstruction. *The bar differential* $D\colon \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}(n) \to \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}(n-1)$ *contracts an internal edge, reducing degree by* 1*. This contraction is* Σ_{n-2}*-equivariant but* not Σ_n*-equivariant: it breaks the symmetry from* Σ_n *to* Σ_{n-2}*, leaking from* $\ker(\mathrm{av}_n)$ *into* $\mathrm{im}(\mathrm{av}_{n-1})$*. No Lie-algebra section can simultaneously respect the fixed-degree splitting and intertwine with* D*.*

(iii) Associator representative. *At degree* 3*, the linearised Drinfeld associator* $[\Omega_{12}, \Omega_{23}] \in \mathrm{End}(V^{\otimes 3})$ *lies entirely in* $\ker(\mathrm{av})$*: it is antisymmetric under* $(1 \leftrightarrow 3)$*, so its* Σ_3*-average vanishes. The infinitesimal braid relation* $[\Omega_{12}, \Omega_{13} + \Omega_{23}] = 0$ *ensures that the commutator is intrinsic, not an artefact of the representation.*

(iv) GRT_1-torsor structure. *The space of dg Lie sections of* av *satisfying the MC equation up to gauge is a torsor for the pro-unipotent Grothendieck–Teichmüller group* GRT_1 *of Drinfeld* [5]*. This is the algebraic shadow of the Etingof–Kazhdan* [6] *non-uniqueness of quantisation: a splitting would canonically reconstruct the quantum group from the modular shadow* κ *alone.*

For Heisenberg ($\dim V = 1$*):* $\ker(\mathrm{av}) = 0$ *at every degree, the extension splits trivially, and no* GRT_1 *ambiguity arises.*

Proof. Part (i): the commutant of Σ_n in $\mathrm{End}(V^{\otimes n})$ is a Lie subalgebra, so $c(A, B) = (1 - \mathrm{av})([A, B]) = 0$.

Part (ii): the partial-trace map $D_{ij}\colon \mathrm{End}(V^{\otimes n}) \to \mathrm{End}(V^{\otimes(n-1)})$ satisfies $D_{ij} \circ \sigma = \sigma|_{\{1,\ldots,n\}\setminus\{j\}} \circ D_{ij}$ only for σ fixing both i and j, i.e. $\Sigma_{n-2} \subset \Sigma_n$. A Σ_n-invariant endomorphism in $\ker(\mathrm{av}_n)$ can map under D to an endomorphism with nonzero av_{n-1} component.

Part (iii): the Σ_3-isotypic decomposition of $[\Omega_{12}, \Omega_{23}]$ consists of sign and standard components only, with zero trivial component (verified in $\mathrm{End}((\mathbb{C}^2)^{\otimes 3})$). The infinitesimal braid relation $[\Omega_{12}, \Omega_{13} + \Omega_{23}] = 0$ ensures the result is intrinsic.

Part (iv) follows from Drinfeld's theorem [5] on the GRT_1-torsor structure of associators, together with the Etingof–Kazhdan quantisation theorem [6]. □

Proposition 4.8 (Non-splitting at genus 1)**.** *The short exact sequence* (4.2) *restricted to genus* 1 *does not split. The three genus-*0 *obstructions persist, and a fourth obstruction arises from the quasi-modular anomaly of the Eisenstein series* E_2*. The genus-*1 *propagator on the elliptic curve* E_τ *is* $\eta^{(1)}_{12} = (\vartheta_1'/\vartheta_1)(z_1 - z_2 \mid \tau)\,(dz_1 - dz_2)$*. The self-energy regularisation produces the Eisenstein series* $E_2(\tau)$*. The averaging map performs the non-holomorphic completion*

$$E_2(\tau) \;\longmapsto\; \widehat{E}_2(\tau) \;:=\; E_2(\tau) - \frac{3}{\pi\ \mathrm{Im}(\tau)}.$$

A section $s\colon \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}(1,1) \to \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}(1,1)$ *would lift* $\widehat{E}_2$ *back to a holomorphic object on the* E_1 *side; the non-holomorphic correction* $-3/(\pi\ \mathrm{Im}(\tau))$ *has no preimage in the holomorphic* E_1 *convolution Lie algebra, so no such section exists.*

Proof. The genus-0 obstructions persist via the embedding of the boundary stratum $\overline{\mathcal{M}}_{0,n+2} \subset \partial\overline{\mathcal{M}}_{1,n}$. The quasi-modular obstruction is independent: it is visible at degree 1 (where Σ_1 is trivial and all genus-0 obstructions vanish), and it concerns the genus-1 fibre (dependence on τ), not the degree (dependence on n). The two obstruction mechanisms are linearly independent in the relevant Ext group. □

Remark 4.9 (The kernel dimension formula)*.* The kernel of av at degree n on the bare tensor power $V^{\otimes n}$ has a closed-form dimension depending only on $d = \dim V$:

$$\dim\bigl(\ker(\mathrm{av}_n)\bigr) \;=\; d^n - \binom{n + d - 1}{d - 1}.$$

The first term $d^n = \dim(V^{\otimes n})$ and the second $\binom{n+d-1}{d-1} = \dim(\mathrm{Sym}^n V)$ is the dimension of the Σ_n-coinvariant. This formula depends only on $\dim V$, not on the Lie-algebraic structure of the representation, by the Schur–Weyl analysis of [14, Proposition 6.1]. At $d = 1$ (Heisenberg): $\ker(\mathrm{av}_n) = 0$ for all n. At $d = 2$ ($\mathfrak{sl}_2$): $\ker(\mathrm{av}_2) = 1$, $\ker(\mathrm{av}_3) = 4$, $\ker(\mathrm{av}_4) = 11$.

5. The E_1 primacy theorem

5.1. Precise statement.

Theorem 5.1 (E_1 primacy). *Let $\mathcal{A}$ be a cyclic chiral algebra on X.*

(i) (Surjectivity.) *The averaging map* $\mathrm{av}\colon \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ *is a surjective dg Lie morphism.*

(ii) (MC projection.) av *sends* $\Theta^{\mathsf{E}_1}_{\mathcal{A}} \mapsto \Theta_{\mathcal{A}}$. *In particular, the shadow obstruction tower* $(\kappa, \mathfrak{C}, \mathfrak{Q}, \ldots)$ *is the degree-by-degree Σ_n-coinvariant image of the R-matrix tower $(r(z), r_3, r_4, \ldots)$.*

(iii) (Bracket preservation.) av *preserves the dg Lie bracket by Σ_n-equivariance of the operad composition maps.*

(iv) (Non-splitting.) *The kernel* $\ker(\mathrm{av})$ *is non-trivial: at degree* 3 *it contains the Drinfeld associator* $\Phi_{\mathrm{KZ}}(\mathcal{A})$*, which is annihilated by* av *but satisfies the pentagon equation in* $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$. *The short exact sequence* $0 \to \ker(\mathrm{av}) \to \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}} \xrightarrow{\mathrm{av}} \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \to 0$ *does not split as dg Lie algebras. At genus* 1*, a fourth obstruction to splitting arises from the quasi-modular anomaly of E_2 (Proposition* 4.8*).*

Proof. Parts (i)–(iii) are established in Theorem 4.2 (surjectivity and bracket preservation as a dg Lie morphism) and Theorem 4.6 (MC projection $\mathrm{av}(\Theta^{\mathsf{E}_1}_{\mathcal{A}}) = \Theta_{\mathcal{A}}$). The map av is the Reynolds operator $\phi \mapsto (1/n!)\sum_\sigma \sigma \cdot \phi$; surjectivity follows because av restricts to the identity on Σ_n-invariant elements, which span the target $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$. The bracket is preserved because operad composition in $F\mathrm{Ass}$ is Σ_n-equivariant: coinvariant projection converts T^c-convolution to Sym^c-convolution (Theorem 3.3).

For (iv), the Drinfeld associator $\Phi_{\mathrm{KZ}}(\mathcal{A}) \in \ker(\mathrm{av})_{0,3}$ satisfies the pentagon equation in $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$ (the genus-0, degree-3 MC equation). Its Σ_3-symmetrisation vanishes because the pentagon identity is antisymmetric under transposition of the two internal edges. Were av split, there would exist a dg Lie section $s\colon \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \hookrightarrow \mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$ with $\mathrm{av} \circ s = \mathrm{id}$. Such a section would lift the commutative MC element $\Theta_{\mathcal{A}}$ to an ordered element in the image of s; but the pentagon constraint on $\ker(\mathrm{av})$ forces the degree-3 component of any lift to involve Φ_{KZ}, which does not lie in the image of any linear section (Proposition 4.7). □

Principle 5.2 (The E_1 primacy thesis). Theorem 5.1 admits a concise restatement. The E_1/ordered structure is the natural primitive of the theory. The E_∞/symmetric/modular structure is its Σ_n-coinvariant shadow under av. The five main theorems A–H are the invariants that survive averaging. Information flows in only one direction: from the rich E_1 side, which contains the R-matrix, the KZ associator, the Drinfeld–Jimbo Yangian, the full braided category of line operators, and the entire spectral scattering data, to the coarser E_∞ side, which contains only the modular characteristic κ, the shadow obstruction tower, and the five structural theorems.

Remark 5.3 (Categorical level of the E_2 braided structure;). The principle above states that the E_1-side contains the R-matrix and the braided category of line operators. To avoid the most common conflation (MP1), we state explicitly *which object carries the* E_2 *structure*: the braided monoidal structure is an E_2-algebra in Cat_∞ on *the Drinfeld center*

$$\mathcal{Z}\big(\mathrm{Rep}^{\mathsf{E}_1}(\mathcal{A})\big),$$

not on $\mathcal{A}$ itself. The algebra $\mathcal{A}$ remains an E_1-object; the E_2 braiding arises one categorical level up, as the universal half-braiding $\sigma_M(N)\colon M \otimes N \to N \otimes M$ constructed from the R-matrix. Quantum groups $U_q(\mathfrak{g})$ are the prototype: $U_q(\mathfrak{g})$ is an E_1-Hopf algebra, and $\mathrm{Rep}(U_q(\mathfrak{g})) = \mathcal{Z}(\mathrm{Rep}(\mathfrak{g}_0)_{\sqrt{\ }})$ is

E_2 in Cat. The Deligne conjecture (classical and chiral) upgrades this to an E_2-action on $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$; the cohomological version is unconditional, the chain-level Tamarkin-type statement is conditional on formality of $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$.

In particular, Dunn additivity $\mathsf{E}_1 \otimes \mathsf{E}_1 = \mathsf{E}_2$ applies to $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ and $\mathrm{Mod}_{\mathcal{A}}$, never to $\mathcal{A}$ itself. Compositions of the form "R-matrix makes $\mathcal{A}$ into an E_2-algebra" are category errors.

5.2. **The five ordered theorems.** The five theorems of modular Koszul duality admit ordered E_1 counterparts, each of which projects to the classical theorem under av.

Theorem 5.4 (Theorem $\mathrm{A}^{\mathsf{E}_1}$: ordered bar-cobar)**.** *The genus-g ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})(g, n)$ is an $F\mathrm{Ass}$-coalgebra with differential satisfying $d^2 = 0$, where $F\mathrm{Ass}$ is the Feynman transform of the ribbon modular operad. The bar and cobar functors form an adjunction*

$$\Omega^{\mathrm{ch},(g)} \dashv \overline{B}^{\mathrm{ord},(g)} : \mathrm{Alg}_{F\mathrm{Ass}} \rightleftarrows \mathrm{CoAlg}_{F\mathrm{Ass}},$$

and coinvariant projection recovers Theorem A of Volume I.

Theorem 5.5 (Theorem $\mathrm{B}^{\mathsf{E}_1}$: ordered inversion)**.** *Assume $\mathcal{A}$ lies on the E_1 Koszul locus, meaning the PBW-completeness hypothesis holds for the ordered bar complex. Then at each genus g, the counit*

$$\Omega^{\mathrm{ch},(g)}\big(\overline{B}^{\mathrm{ord}}(\mathcal{A})(g, \bullet)\big) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism (inversion, not Koszul duality), and the E_1 linear Koszul dual $\mathcal{A}^{!,\mathsf{E}_1}$ obtained by linear duality on a finite-dimensional model of $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ is an E_1-chiral algebra at each such genus.

Theorem 5.6 (Theorem $\mathrm{C}^{\mathsf{E}_1}$: ordered complementarity)**.** *At each genus g and degree n with $2g - 2 + n > 0$, the ordered complementarity takes the form*

$$Q^{\mathsf{E}_1}_{g,n}(\mathcal{A}) + Q^{\mathsf{E}_1}_{g,n}(\mathcal{A}^{!,\mathsf{E}_1}) \;=\; H^*\big(\overline{\mathcal{M}}^{\mathrm{rib}}_{g,n},\, Z^{\mathsf{E}_1}(\mathcal{A})\big),$$

where $Z^{\mathsf{E}_1}(\mathcal{A})$ is the E_1 centre, the space of R-matrix-equivariant elements. The scalar identity $\kappa + \kappa^! = 0$ for Kac–Moody and free-field families (and $\kappa + \kappa^! = 13$ for Virasoro) is the Σ_n-coinvariant image.

Theorem 5.7 (Theorem $\mathrm{D}^{\mathsf{E}_1}$: ordered degree-2 package)**.** *The formal ordered degree-2 shadow series*

$$R^{\mathsf{E}_1,\mathrm{bin}}(z;\ \ ) \;=\; \sum_{g \geq 0} \ {}^{2g}\, r_g(z)$$

obtained from the genus-refined degree-2 projection of the E_1 Maurer–Cartan element is the E_1 degree-2 characteristic package of $\mathcal{A}$. It is universal, additive, antisymmetric under opposite-duality, and satisfies $\mathrm{av}(r_0(z)) = \kappa(\mathcal{A})$ *on abelian families, while for non-abelian affine Kac–Moody* $\mathrm{av}(r_0(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$ *and $\kappa(\mathcal{A}) = \kappa_{\mathrm{dp}}(\mathcal{A}) + \dim(\mathfrak{g})/2$. The coinvariant recovery* $\mathrm{av}(R^{\mathsf{E}_1,\mathrm{bin}}) = \sum\ {}^{2g} F^{(2)}_g$ *is the scalar genus-2 modular characteristic series of Theorem D.*

Theorem 5.8 (Theorem $\mathrm{H}^{\mathsf{E}_1}$: ordered chiral Hochschild)**.** *For every genus g and complete $\mathcal{A}$-bimodule M, the genus-g ordered chiral Hochschild homology is computed coalgebraically by the genus-g ordered chiral coHochschild complex of the ordered bar coalgebra $C = \overline{B}^{\mathrm{ord}}(\mathcal{A})$, with twisted differential involving the curvature term $\kappa(\mathcal{A}) \cdot \lambda_g$ in the uniform-weight case* (UNIFORM-WEIGHT). *At genus $g \geq 2$ with multi-weight field content the scalar formula receives a cross-channel correction $\delta F^{\mathrm{cross}}_g$* (ALL-WEIGHT $+\delta F^{\mathrm{cross}}_g$)*. The Σ_n-coinvariant recovers the symmetric chiral Hochschild complex of Theorem H.*

Remark 5.9 (Theorem D is the cleanest example). Among the five, Theorem D makes the E_1 primacy thesis most concrete. The scalar modular characteristic κ is a single number per family, and the family formulas are distinct: for $\mathcal{W}_N$ the harmonic number is $H_N = \sum_{j=1}^{N} 1/j$ and must not be copied from a different family. Its degree-2 lift is a meromorphic function of a single variable, the collision coordinate z. The statement $\mathrm{av}(r(z)) = \kappa$ says this exactly for Heisenberg, while for non-abelian affine Kac–Moody the same averaging recovers only κ_{dp} and the full κ adds the Sugawara shift $\dim(\mathfrak{g})/2$. The scalar shadow is still the Σ_2-average of a function. This is a strict projection; the full function $r(z)$ carries information about the braiding, the quantum group, and the line-operator algebra that no scalar invariant can see.

Remark 5.10 (What averaging forgets). The kernel of av contains several objects of independent interest. At degree 2, $\ker(\mathrm{av})$ is the antisymmetric part of $r(z)$; for bosonic generators of even desuspended degree this vanishes identically, and κ recovers all of $r(z)$. For generators of odd desuspended degree, the kernel is nontrivial and captures the parity-dependent Eulerian weight structure on the bar complex. At degree 3, $\ker(\mathrm{av})$ contains the linearised Drinfeld commutator $[\Omega_{12}, \Omega_{23}]$ and, by the non-splitting proposition, obstructs a canonical reconstruction of the full associator from $\mathfrak{C}$. The GRT_1-torsor structure of the splittings of av is the operadic shadow of Etingof–Kazhdan non-uniqueness of quantisation.

6. The formality bridge: E_∞ input

The ordered bar complex is defined for every E_1-chiral algebra, but the symmetric bar exists only when the $\mathcal{D}_X$-module descends to the unordered Ran space. For E_∞-chiral input the descent holds, and the question is: does the coinvariant projection lose information? The answer at the cohomological level is no; at the chain level the loss is precisely the R-matrix and the associator.

Remark 6.1 (Formality for E_∞-chiral input). Every standard family in conformal field theory (Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}$-algebras, lattice vertex algebras, $\beta\gamma$ systems, *bc* ghosts, Bershadsky–Polyakov) is E_∞-chiral: the OPE is symmetric in the operator ordering up to signs from the Arnold form, and the $\mathcal{D}_X$-module structure descends to the symmetric Ran space. For such algebras the E_1 refinement adds no new data at the level of bar *cohomology*: the natural comparison map

$$H^*(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \xrightarrow{\sim} H^*(\overline{B}^{\Sigma}(\mathcal{A}))$$

is an isomorphism. The ordered bar carries additional data at the *chain level*: the R-matrix, the KZ connection, and the Drinfeld associator are chain-level objects that project to zero in cohomology but determine the braided monoidal structure on the category of $\mathcal{A}$-modules.

6.1. The formality theorem.

Theorem 6.2 (Formality bridge for E_∞-chiral algebras). *Let $\mathcal{A}$ be an E_∞-chiral algebra on X and Φ a Drinfeld associator. The Kontsevich–Tamarkin operad formality quasi-isomorphism induces a quasi-isomorphism of coalgebras*

$$\overline{B}^{\mathrm{ord}}(\mathcal{A}) \xrightarrow[\Phi]{\sim} \overline{B}^{\Sigma}(\mathcal{A}) \tag{6.1}$$

that intertwines the deconcatenation coproduct on $T^c(s^{-1}\bar{\mathcal{A}})$ with the coshuffle coproduct on $\mathrm{Sym}^c(s^{-1}\bar{\mathcal{A}})$ at the cohomological level.

In particular, the coinvariant projection $\pi_{\Sigma_n} \colon (s^{-1}\bar{\mathcal{A}})^{\otimes n} \twoheadrightarrow (s^{-1}\bar{\mathcal{A}})^{\odot n}$ induces an isomorphism on bar cohomology:

$$H^*\big(\overline{B}^{\mathrm{ord}}(\mathcal{A})\big) \xrightarrow{\cong} H^*\big(\overline{B}^{\Sigma}(\mathcal{A})\big).$$

The transferred A_∞-structure on the cohomological model $H^*(\mathcal{A})$ may carry non-trivial higher operations m_k^{tr} for $k \geq 3$; what the formality bridge guarantees is that ordered and symmetric complexes compute the same cohomology.

Proof. The argument proceeds in four steps.

Step 1: Koszul resolution of the Σ_n-coinvariant. Working in characteristic zero, the group algebra $k[\Sigma_n]$ is semisimple (Maschke's theorem). The idempotent

$$e_n \;:=\; \frac{1}{n!}\sum_{\sigma\in\Sigma_n}\sigma \;\in\; k[\Sigma_n]$$

is the Reynolds operator projecting $V^{\otimes n}$ onto $\mathrm{Sym}^n(V) = (V^{\otimes n})_{\Sigma_n}$, and the complement $(1-e_n)(V^{\otimes n})$ is the kernel of av_n. Semisimplicity gives a direct sum decomposition

$$V^{\otimes n} \;\cong\; \mathrm{Sym}^n(V) \;\oplus\; \ker(\mathrm{av}_n) \tag{6.2}$$

as $k[\Sigma_n]$-modules. This decomposition is compatible with the bar differential by the Σ_n-equivariance of the OPE for E_∞-chiral algebras.

Step 2: Group cohomology vanishing. In characteristic zero,

$$H^i(\Sigma_n,\, M) \;=\; 0 \qquad \text{for all } i>0 \text{ and all } k[\Sigma_n]\text{-modules } M. \tag{6.3}$$

This vanishing follows from semisimplicity: the category $k[\Sigma_n]$-mod has global dimension zero, so $\mathrm{Ext}^i_{k[\Sigma_n]}(k, M) = 0$ for $i>0$. The vanishing ensures that the coinvariant projection $\pi_{\Sigma_n}\colon \overline{B}^{\mathrm{ord}}(\mathcal{A}) \twoheadrightarrow \overline{B}^{\Sigma}(\mathcal{A})$ is exact: no higher Σ_n-cohomology obstructs the passage from ordered to symmetric.

Step 3: The quasi-isomorphism. Consider the coinvariant short exact sequence at each degree n:

$$0 \;\to\; \ker(\mathrm{av}_n) \;\to\; (s^{-1}\bar{\mathcal{A}})^{\otimes n} \;\xrightarrow{\pi_{\Sigma_n}}\; (s^{-1}\bar{\mathcal{A}})^{\odot n} \;\to\; 0.$$

The bar differential d_{bar} preserves the decomposition (6.2) because the E_∞-chiral OPE is Σ_n-equivariant: the collision residue $a_i \star a_{i+1}$ is symmetric in a_i, a_{i+1} up to the graded sign from the Arnold form. The decomposition (6.2) at the chain level, combined with the cohomology vanishing of (6.3), gives

$$H^*\big(\ker(\mathrm{av}_n), d_{\mathrm{bar}}\big) \;\hookrightarrow\; H^*\big(\Sigma_n, (s^{-1}\bar{\mathcal{A}})^{\otimes n}\big) \;=\; 0 \quad \text{for } *>0.$$

The long exact sequence in cohomology then forces π_{Σ_n} to be a quasi-isomorphism.

Step 4: Explicit quasi-isomorphism map. The Kontsevich–Tamarkin operad formality (Kontsevich [13], Tamarkin, see also [15]) provides a quasi-isomorphism

$$\varphi_\Phi\colon C_*(\overline{\mathrm{FM}}_n(\mathbb{C})) \;\xrightarrow{\sim}\; H^*(\overline{\mathrm{FM}}_n(\mathbb{C}))$$

depending on the Drinfeld associator Φ. Tensoring with $(s^{-1}\bar{\mathcal{A}})^{\otimes n}$ and taking R-twisted Σ_n-coinvariants (equivalently $\overline{B}^{\Sigma}_n(\mathcal{A}) \simeq (\overline{B}^{\mathrm{ord}}_n(\mathcal{A}))^{R\text{-}\Sigma_n\text{-coinv}}$; the twist reduces to the naive one on the pole-free subclass;) gives the quasi-isomorphism (6.1). The dependence on Φ is a GRT_1-ambiguity: different choices of associator yield homotopic maps. □

Remark 6.3 (What the bridge preserves and what it loses)*.* The formality bridge preserves the bar *cohomology*: homological invariants such as the dimension of the ordered bar cohomology groups, the Koszul property (PBW-degree concentration), and the five theorem statements. It loses chain-level data: the R-matrix $r(z)$ is a chain-level datum on the ordered side that becomes trivial (Σ_2-average $\to \kappa$) on the symmetric side. The Drinfeld associator Φ_{KZ} is a chain-level datum at degree 3 that maps to zero under av. The braided monoidal structure on the category of $\mathcal{A}$-modules is encoded in the chain-level ordered bar complex and is invisible from the symmetric bar cohomology.

7. The formality failure: E_1 input

For genuinely E_1-chiral input (Yangians, Etingof–Kazhdan quantum vertex algebras), the formality bridge of Section 6 breaks down. The obstruction is clean: the $\mathcal{D}_X$-module underlying the OPE does not descend to the symmetric quotient $X^{(n)}$, because the R-matrix $S(z) \neq \mathrm{id}$ breaks the Σ_n-equivariance of the bar differential. The ordered bar is the only bar that exists, and the averaging map has no target.

7.1. The theorem.

Theorem 7.1 (Formality failure for genuinely E_1-chiral algebras). *Let $\mathcal{A}$ be a genuinely E_1-chiral algebra (tier (c) of the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy: the vertex R-matrix $S(z) \neq \mathrm{id}$ is not derived from any E_∞-chiral OPE). The formality bridge of Theorem 6.2 fails in the following precise sense.*

(i) (Operad formality is unconditional.) *The Kontsevich–Tamarkin formality of $C_*(\overline{\mathrm{FM}}_n(\mathbb{C}))$ as an operad depends only on the operad structure, not on the specific algebra. It holds for E_1-chiral input.*

(ii) (Equivariance fails: the $\mathcal{D}$-module does not descend.) *The factorization $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}$ on $\mathrm{Conf}_n^{\mathrm{ord}}(X)$ is* not *Σ_n-equivariant. The obstruction is the R-matrix: the monodromy of $\mathcal{F}_n^{\mathrm{ord}}$ around the diagonal Δ_{ij} is*

$$\mathrm{Mon}_{\gamma_{ij}}(\mathcal{F}_n^{\mathrm{ord}}) \;=\; S_{ij}(z) \;\neq\; \mathrm{id},$$

where $S_{ij}(z)$ is the braiding by the R-matrix on the i-th and j-th tensor factors. Since $S_{ij} \neq \mathrm{id}$, the $\mathcal{D}$-module does not descend from $\mathrm{Conf}_n^{\mathrm{ord}}(X)$ to the symmetric quotient $\mathrm{Conf}_n(X)/\Sigma_n$; equivalently, the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ does not exist as a factorization coalgebra on the unordered Ran space $\mathrm{Ran}(X)$.

(iii) (Dimension count: the Arnold algebra.) *By Arnold's theorem* [1], *the cohomology algebra $H^*(\mathrm{Conf}_n(\mathbb{C}))$ is the quotient of the exterior algebra on generators $\omega_{ij} = [d\log(z_i - z_j)]$ of degree 1 ($1 \le i < j \le n$) by the Arnold relation $\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij} = 0$. Its Poincaré polynomial is*

$$P_n(t) \;=\; \prod_{j=0}^{n-1}(1 + jt).$$

Setting $t = 1$ gives $P_n(1) = n!$ as total dimension. At $n = 2$: $P_2(1) = 2$; at $n = 3$: $P_3(1) = 6$; at $n = 4$: $P_4(1) = 24$. The ordered chiral chain complex at degree n has dimension

$$\dim C_n^{\mathrm{ord}}(D^\times, \mathcal{A}) \;=\; (\dim \bar{\mathcal{A}})^n \cdot n!,$$

where the factor $n!$ from the Arnold algebra measures the holomorphic data carried by the ordered complex beyond the topological Hochschild complex (which has dimension $(\dim \mathcal{A})^n$, without the $n!$ factor).

(iv) (Braid group representation.) *The fundamental group $\pi_1(\mathrm{Conf}_n(\mathbb{C})) = \mathrm{PBr}_n$ is the pure braid group on n strands. The $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}$ gives a representation*

$$\rho_n\colon\ \mathrm{PBr}_n \;\longrightarrow\; \mathrm{Aut}\big(C_n^{\mathrm{ord}}(D^\times, \mathcal{A})\big) \tag{7.1}$$

by monodromy. By the Drinfeld–Kohno theorem, the local monodromy around Δ_{ij} is $\exp(2\pi i \cdot r_{ij})$, where r_{ij} is the classical r-matrix acting on the i-th and j-th factors. For E_∞ input, r_{ij} is scalar and the braid group acts through Σ_n (the abelianisation); for genuinely E_1 input, the braid representation is faithful and the ordered chiral homology carries the full braided monoidal structure.

Proof. Part (i): Kontsevich–Tamarkin operad formality depends on the structure maps of the Fulton–MacPherson operad, not on the algebra to which it is applied. The formality map φ_Φ of (6.1) exists unconditionally.

Part (ii): the factorization $\mathcal{D}$-module $\mathcal{F}_n^{\mathrm{ord}}$ on $\mathrm{Conf}_n^{\mathrm{ord}}(X)$ is constructed from the OPE of $\mathcal{A}$. The OPE $Y(a,z)b$ and $Y(b,-z)a$ are related by the R-matrix: $Y(a,z)b = S(z)\cdot Y(b,-z)a$. If $S(z) \neq \mathrm{id}$, the transposition $\sigma_{ij}\colon (z_i, z_j) \mapsto (z_j, z_i)$ acts on $\mathcal{F}_n^{\mathrm{ord}}$ by $S_{ij}(z_i - z_j) \neq \mathrm{id}$, and the $\mathcal{D}$-module does not descend to $X^{(n)}$. The ordered complex retains holomorphic data from the spectral-parameter dependence of $S(z)$ that has no counterpart in the cyclic bar complex.

Part (iii): the Poincaré polynomial $P_n(t) = \prod_{j=0}^{n-1}(1 + jt)$ follows from Arnold's computation of $H^*(\mathrm{Conf}_n(\mathbb{C}))$ [1]: the generators ω_{ij} are of degree 1, the Arnold relation kills the excess, and the Hilbert function is determined by the recursion $P_n(t) = (1 + (n-1)t) \cdot P_{n-1}(t)$ with $P_1(t) = 1$.

Part (iv): the Drinfeld–Kohno theorem (Drinfeld [5], Kohno [12]) identifies the monodromy representation of the KZ connection with the quantum group R-matrix action, giving the braid representation (7.1). For E_∞ input, the R-matrix is the Koszul-signed identity $S(z) = (-1)^{|a||b|}\mathrm{id}$, so the braid group acts through signs. For genuinely E_1 input, the representation is faithful: the Yang–Baxter equation on $r(z)$ ensures consistency of the braid relations, and non-triviality of $S(z)$ ensures non-triviality of the representation. □

7.2. The Yangian as paradigm.

Remark 7.2 ($n!$ versus 1)*.* The dimension count exposes the gap: the E_1 convolution space $\mathrm{Hom}(\mathcal{M}_{\mathrm{Ass}}(g,n), \mathrm{End}_{\mathcal{A}}(n))$ has dimension $n! \cdot \dim\big(\mathrm{Hom}_{\Sigma_n}(\mathcal{M}_{\mathrm{Com}}(g,n), \mathrm{End}_{\mathcal{A}}(n))\big)$ at each degree, by the multiplicity of the regular representation in the coinvariant decomposition. For E_∞ input the $n!$ factor is redundant: the data is Σ_n-equivariant and the coinvariant projection is a quasi-isomorphism. For E_1 input the $n!$ factor is essential: the Maurer–Cartan element $\Theta_{\mathcal{A}}^{\mathsf{E}_1}$ uses all $n!$ orderings, the R-matrix cannot be recovered from its Σ_2-coinvariant, and the symmetric bar $\overline{B}^{\Sigma}(\mathcal{A})$ does not exist as a factorization coalgebra on the *unordered* Ran space (the $\mathcal{D}$-module does not descend to $X^{(n)}$).

The paradigmatic example is the Yangian $Y\ (\mathfrak{g})$. Its R-matrix $r_{\mathrm{YB}}(z) = \ \Omega_{\mathfrak{g}}/z$ is a solution of the classical Yang–Baxter equation, and the bar complex $\overline{B}^{\mathrm{ord}}(Y\ (\mathfrak{g}))$ is an $F\mathrm{Ass}$-coalgebra whose Koszul dual is the affine Kac–Moody algebra. The symmetric bar $\overline{B}^{\Sigma}(Y\ (\mathfrak{g}))$ does not exist because the Yangian's OPE is not symmetric: the braiding $S(z) = 1 + \ P/z$ satisfies $S(z) \neq S(-z)$. This is the cleanest demonstration of E_1 primacy: the ordered bar is the only bar that can be formed, and the averaging map has no target.

Definition 7.3 (Ordered Koszul chiral algebra)**.** An E_1-chiral algebra $\mathcal{A}$ is *ordered Koszul* if (i) the universal twisting morphism $\tau_{\mathcal{A}}^{\mathrm{ord}} \colon \overline{B}^{\mathrm{ord}}(\mathcal{A}) \to \mathcal{A}$ is a chiral Koszul morphism in the ordered sense (the bar–cobar counit $\Omega^{\mathrm{ord}}(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism of E_1-chiral algebras); and (ii) the PBW spectral sequence of $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ collapses at E_2 in the ordered grading.

Proposition 7.4 (The Yangian is ordered Koszul)**.** *For every finite-dimensional simple Lie algebra* $\mathfrak{g}$ *and every* $\ \neq 0$*, the Yangian* $Y\ (\mathfrak{g})^{\mathrm{ch}}$ *is ordered Koszul in the sense of Definition* 7.3*.*

Proof sketch. Present $Y\ (\mathfrak{g})^{\mathrm{ch}}$ in Drinfeld's second form as generated by $\{\xi_{i,r}^{\pm}, h_{i,r}\}$ modulo the loop–algebra relations; by Etingof–Kazhdan flatness [?EK98] the resulting filtration is PBW-flat over . Koszulity of the pure-braid Orlik–Solomon algebra (Shelton–Yuzvinsky [?SheltonYuzvinsky]) identifies the ordered bar cohomology with the quadratic dual under the Drinfeld–KZ bijection; E_2-collapse is the Etingof–Kazhdan–Drinfeld flatness statement. The EK monodromy provides the counit quasi-isomorphism witness at generic . The symmetric bar has no target, so the ordered presentation is the unique one. □

Lemma 7.5 (R-twisted Σ_n descent (unitary R))**.** *Let* $\mathcal{A}$ *be an ordered Koszul* E_1*-chiral algebra with R-matrix* $R(z) = 1 + \ r(z) + O(\ ^2)$ *satisfying the classical Yang–Baxter equation* and *unitarity*

$R(z)\,R^{op}(-z) = \mathrm{id}$, *and let* $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ *be its ordered bar complex. The twisted* Σ_n*-action on* $\overline{B}^{\mathrm{ord}}(\mathcal{A})_n$*, defined on a transposition* $\sigma_{i,i+1}$ *by* $\sigma_{i,i+1}\cdot(a_1\otimes\cdots\otimes a_n) = R_{i,i+1}(z_i - z_{i+1})\cdot(a_1\otimes\cdots\otimes a_{i+1}\otimes a_i\otimes\cdots\otimes a_n)$*, extends to an action of the pure braid group* PBr_n *by Yang–Baxter; unitarity implements the transposition relation* $\sigma_{i,i+1}^2 = \mathrm{id}$ *at the monodromy level, descending the action to* Σ_n*. The quotient* $\overline{B}^{\mathrm{ord}}(\mathcal{A})_n/\Sigma_n$ *under this R-twist is the symmetric bar* $\overline{B}^{\Sigma}(\mathcal{A})_n$ *whenever the latter exists* (*i.e. on the* E_∞*-factorization locus*)*. The lossy descent map* $\mathrm{av}\colon \overline{B}^{\mathrm{ord}}(\mathcal{A}) \to \overline{B}^{\Sigma}(\mathcal{A})$ *is precisely the R-twisted Reynolds projector associated with this action. Unitarity is automatic for rational Yangians* $Y(\mathfrak{g})$ *and for trigonometric* $U_q(\widehat{\mathfrak{g}})$ *at generic* q*, vacuous for* $R(z) = \tau$*, convention-dependent for elliptic Belavin/Felder R-matrices (cf. Vol I, Lemma* `lem:R-twisted-descent` *and its scope remark).*

This lemma is the E_1 analogue of the classical Dunn–Lurie E_∞-descent and records that the gap between $\overline{B}^{\mathrm{ord}}$ and $\overline{B}^{\Sigma}$ is controlled not by an abelian Σ_n-action but by a braid representation.

Example 7.6 (Yangian $Y(\mathfrak{sl}_2)$: the braid representation)**.** For $\mathfrak{g} = \mathfrak{sl}_2$ and $V = \mathbb{C}^2$ the fundamental representation, the R-matrix in the fundamental representation is $R(u) = u\,I_4 + \quad P \in \mathrm{End}(\mathbb{C}^2\otimes\mathbb{C}^2)$, where $P(e_i\otimes e_j) = e_j\otimes e_i$. The braid group $\mathrm{PBr}_2 = \mathbb{Z}$ acts on $C_2^{\mathrm{ord}}(D^\times, Y(\mathfrak{sl}_2))$ by the monodromy operator

$$\rho_2(\gamma_{12}) \;=\; \exp(2\pi i\cdot r_{12}) \;=\; \exp\!\big(2\pi i \quad \Omega_{\mathfrak{sl}_2}/z\big),$$

which at $z = 1$ and generic   has four distinct eigenvalues on $V\otimes V = V_0\oplus V_1$ (the singlet and triplet of $\mathfrak{sl}_2$). The symmetric bar would project this to the Σ_2-coinvariant $\mathrm{Sym}^2(V) = V_1$, losing the singlet channel; the ordered bar retains both isospin channels.

8. Heisenberg and Kac–Moody: complete E_1 computation

This section gives the complete degree-2 ordered computation for the two simplest families and records the passage through av to the scalar shadow. The computations are self-contained and illustrate every step of the E_1 primacy thesis.

8.1. Heisenberg.

For $\mathcal{H}_k$ with OPE $J(z)J(w)\sim k/(z-w)^2$:

- $r(z) = k/z$ (equation (2.1)).
- $\mathrm{av}(r(z)) = \kappa(\mathcal{H}_k) = k$ (the Σ_2-average of k/z against $d\log(z)$ extracts the residue k, which is the full modular characteristic).
- $\ker(\mathrm{av}_2) = 0$ (one-dimensional representation).
- Bar cohomology $H^{-1}(\overline{B}^{\mathrm{ord}}(\mathcal{H}_k)) = \mathbb{C}\cdot s^{-1}J$: concentrated in bar degree 1. Koszul dual: $\mathcal{H}_k^! = (\mathcal{H}_k^{\mathrm{i}})^\vee$ with $\kappa(\mathcal{H}_k^!) = -k$. Complementarity: $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = k + (-k) = 0$ (Kac–Moody and free-field families have $\kappa+\kappa^! = 0$; Virasoro has $\kappa+\kappa^! = 13$; the value is family-dependent).
- The ordered Yangian is trivial: $Y(\mathbb{C})\cong\mathbb{C}[\ \ ]$. The KZ connection is scalar (no braiding). The GRT_1 ambiguity is absent.

The Heisenberg computation makes the passage through av explicit. The degree-2 ordered MC component is

$$\Theta^{\mathsf{E}_1,2}_{\mathcal{H}_k} \;=\; k\cdot[s^{-1}J|s^{-1}J]\otimes\frac{dz}{z} \;\in\; (s^{-1}\bar{\mathcal{H}}_k)^{\otimes 2}\otimes\Omega^1(\mathrm{Conf}_2^{\mathrm{ord}}(\mathbb{C})),$$

a single generator tensored with the Arnold form $\eta = dz/z$ on the two-point ordered configuration. The averaging map acts by the residue extraction:

$$\mathrm{av}(\Theta^{\mathsf{E}_1,2}_{\mathcal{H}_k}) \;=\; \frac{1}{2\pi i}\oint_{\gamma_0}\frac{k}{z}\,dz \;=\; k \;=\; \kappa(\mathcal{H}_k).$$

The kernel $\ker(\mathrm{av}_2) = 0$ because $\dim V = 1$, so the ordered and symmetric bars carry identical degree-2 data. At degree 3 and higher, the Arnold relation is automatically satisfied (no non-abelian Casimirs exist), and the Heisenberg shadow tower terminates at $S_2 = \kappa = k$: this is class G, with $\Delta = 8\kappa \cdot S_4 = 0$ (since $S_4 = 0$).

The KZ connection at n points is scalar:

$$\nabla^{\mathcal{H}_k}_{\mathrm{KZ}} \;=\; d - k \sum_{i<j} \frac{dz_{ij}}{z_i - z_j},$$

and its monodromy is abelian: the representation $\rho_n\colon\ \mathrm{PBr}_n \longrightarrow \mathbb{C}^\times$ factors through $\mathrm{PBr}_n \twoheadrightarrow \mathbb{Z}^{\binom{n}{2}}$, and each generator γ_{ij} acts by the scalar $e^{2\pi i k}$. No Drinfeld associator appears.

8.2. **Affine Kac–Moody.** For $\widehat{\mathfrak{g}}_k$ with $\mathfrak{g}$ simple, $\dim(\mathfrak{g}) = d$, $h^\vee$ the dual Coxeter number:

- $r(z) = k\,\Omega_{\mathfrak{g}}/z$ (equation (2.2); trace-form convention; at $k = 0$ the r-matrix vanishes identically).
- $\mathrm{av}(r(z)) = \kappa_{\mathrm{dp}} = k\,d/(2h^\vee)$ (double-pole channel).
- $\kappa(\widehat{\mathfrak{g}}_k) = d(k + h^\vee)/(2h^\vee)$ (equation (4.1); at $k = 0$: $\kappa = d/2$, not zero; at $k = -h^\vee$: $\kappa = 0$).
- $\ker(\mathrm{av}_2) = (d^2 - d(d+1)/2)$-dimensional: the antisymmetric part of $\Omega_{\mathfrak{g}}$ in $\mathfrak{g} \otimes \mathfrak{g}$.

The degree-2 ordered MC component is

$$\Theta^{\mathsf{E}_1,2}_{\widehat{\mathfrak{g}}_k} \;=\; \sum_a k\, t^a \otimes t_a \;\cdot\; [s^{-1}J^a | s^{-1}J^b] \otimes \frac{dz}{z} \;\in\; (s^{-1}\bar{V})^{\otimes 2} \otimes \Omega^1(\mathrm{Conf}^{\mathrm{ord}}_2(\mathbb{C})),$$

where $\{t^a\}$ is a basis of $\mathfrak{g}$ with dual basis $\{t_a\}$ under the Killing form, and V denotes the vacuum module of $\widehat{\mathfrak{g}}_k$. The Σ_2-average extracts the trace over $\mathfrak{g}$:

$$\mathrm{av}(\Theta^{\mathsf{E}_1,2}_{\widehat{\mathfrak{g}}_k}) \;=\; \frac{1}{2\pi i}\oint_{\gamma_0} k\,\frac{\mathrm{tr}(\Omega_{\mathfrak{g}})}{z}\,dz \;=\; k \cdot \frac{d}{2h^\vee} \;=\; \kappa_{\mathrm{dp}},$$

where $\mathrm{tr}(\Omega_{\mathfrak{g}}) = \sum_a \kappa^{ab}\kappa_{ab} = d/(2h^\vee)$ in the normalisation where $\kappa(t^a, t_a) = 1$. The full modular characteristic adds the Sugawara simple-pole self-contraction:

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}} + \frac{d}{2} \;=\; \frac{d(k+h^\vee)}{2h^\vee}.$$

The extra $d/2$ arises from the adjoint Casimir eigenvalue $2h^\vee$: the simple-pole OPE $f^{ab}{}_c J^c(w)/(z-w)$ contributes through the self-contraction $\sum_{a,c} f^{ac}{}_d f^{ad}{}_c = 2h^\vee \cdot \delta^{cd}$, which after normalisation yields $d/2$.

8.2.1. *The KZ connection and its monodromy.* The n-point KZ connection on the punctured disk is

$$\nabla_{\mathrm{KZ}} \;=\; d \;-\; \sum_{1 \le i < j \le n} \frac{k\,\Omega_{ij}}{z_i - z_j}\,dz_{ij},$$

where $\Omega_{ij} = \sum_a t^a_i \otimes t_{a,j}$ acts on the i-th and j-th tensor factors of $V^{\otimes n}$, and $dz_{ij} = dz_i - dz_j$ (not $d\log(z_i - z_j)$). The connection is flat: its curvature vanishes by the infinitesimal braid relation

$$[\Omega_{ij},\, \Omega_{ik} + \Omega_{jk}] \;=\; 0 \qquad \text{for distinct } i, j, k. \tag{8.1}$$

The monodromy of ∇_{KZ} at $n = 2$ on the configuration space $\mathrm{Conf}_2(\mathbb{C})$ is the single operator

$$\mathrm{Mon}(\nabla_{\mathrm{KZ}}) \;=\; \exp\!\big(2\pi i \cdot k\,\Omega_{\mathfrak{g}}\big) \;\in\; \mathrm{Aut}(V \otimes V).$$

8.2.2. *The Drinfeld associator and pentagon.* At $n = 3$, the monodromy of ∇_{KZ} on $\mathrm{Conf}_3(\mathbb{P}^1 \setminus \{0, 1, \infty\})$ produces the Drinfeld associator

$$\Phi_{\mathrm{KZ}} = \Phi(k\,\Omega_{12},\, k\,\Omega_{23}) \in \exp(\hat{\mathfrak{t}}_3) \subset \mathrm{End}(V^{\otimes 3}),$$

where $\hat{\mathfrak{t}}_3$ is the completed Drinfeld–Kohno Lie algebra generated by Ω_{12} and Ω_{23} subject to the infinitesimal braid relation (8.1). The associator Φ_{KZ} satisfies the pentagon equation in $\mathfrak{g}^{\mathsf{E}_1}_{\mathcal{A}}$:

$$\Phi_{1,2,34}\,\Phi_{12,3,4} = \Phi_{2,3,4}\,\Phi_{1,23,4}\,\Phi_{1,2,3},$$

where $\Phi_{i,j,k}$ denotes the associator acting on the (i, j, k) tensor factors. This pentagon is the degree-3 MC equation on the ordered side (the five faces of the Stasheff associahedron K_4).

The associator projects to zero under av: the linearised commutator $[\Omega_{12}, \Omega_{23}]$ is antisymmetric under $(1 \leftrightarrow 3)$, so $\mathrm{av}_3(\Phi_{\mathrm{KZ}}) = 0$. The pentagon constraint survives averaging as the trivial identity $\mathfrak{C} = 0$ on the symmetric shadow, reflecting the fact that no Σ_3-coinvariant component of the pentagon carries content.

8.2.3. *Worked example:* $\mathfrak{g} = \mathfrak{sl}_2$. For $\mathfrak{g} = \mathfrak{sl}_2$ we have $d = 3$ and $h^\vee = 2$. The modular characteristic is

$$\kappa(\widehat{\mathfrak{sl}}_2,\, k) = \frac{3(k+2)}{4}.$$

At $k = 0$: $\kappa = 3/2 = d/2$. At $k = -2$ (critical): $\kappa = 0$.

The Yang R-matrix in the 2-dimensional fundamental representation $V = \mathbb{C}^2$ is

$$R(u) = u\,I_4 + \quad P \in \mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2), \qquad = \frac{1}{k+2},$$

where P is the permutation matrix. In the standard basis $\{e_1 \otimes e_1,\ e_1 \otimes e_2,\ e_2 \otimes e_1,\ e_2 \otimes e_2\}$, this is the 4×4 matrix

$$R(u) = \begin{pmatrix} u + & 0 & 0 & 0 \\ 0 & u & & 0 \\ 0 & & u & 0 \\ 0 & 0 & 0 & u + \end{pmatrix}.$$

The Yang–Baxter equation $R_{12}(u-v)\,R_{13}(u)\,R_{23}(v) = R_{23}(v)\,R_{13}(u)\,R_{12}(u-v)$ is verified by direct computation on $(\mathbb{C}^2)^{\otimes 3}$. The unitarity relation is $R(u)\,R(-u) = (u^2 - \ {}^2)\,I_4$.

The RTT relation $R_{12}(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R_{12}(u-v)$ at level 1 gives the commutation relations of the Yangian $Y\ (\mathfrak{sl}_2)$:

$$[t^{(1)}_{ij},\, t^{(1)}_{kl}] = \quad \big(\delta_{kj}\,t^{(1)}_{il} - \delta_{il}\,t^{(1)}_{kj}\big),$$

which is $\mathfrak{sl}_2$ at $\ \neq 0$ (recovering the ordinary Lie bracket $[e_{ij}, e_{kl}] = \delta_{kj}e_{il} - \delta_{il}e_{kj}$), and becomes abelian at $\ = 0$.

The KZ monodromy on $\mathrm{Conf}_2(\mathbb{C})$ decomposes $V \otimes V = \mathbb{C}^2 \otimes \mathbb{C}^2$ into the singlet V_0 (dimension 1) and the triplet V_1 (dimension 3), with eigenvalues

$$\exp(2\pi i \cdot k\,\Omega_{\mathfrak{sl}_2}) = \begin{cases} e^{-3\pi i k/2} & \text{on } V_0, \\ e^{\pi i k/2} & \text{on } V_1. \end{cases}$$

At integer level $k \in \mathbb{Z}_{\geq 1}$, these are roots of unity, and the representation category of $U_q(\mathfrak{sl}_2)$ at $q = e^{\pi i/(k+2)}$ is semisimple with $k + 1$ integrable modules $V_0, \ldots, V_k$.

9. The chiral quantum group equivalence

On the Koszul locus, three structures on an E_1-chiral algebra determine each other: the R-matrix, the A_∞ structure, and the spectral coproduct. This section records the equivalence theorem, extends it to the $\mathfrak{gl}_N$ tower for all $N \geq 1$, exhibits the universal Miura coefficient, and recovers the Verlinde formula from ordered chiral homology.

9.1. The equivalence theorem.

Theorem 9.1 (Chiral quantum group equivalence)**.** *On the Koszul locus, the following three structures on an E_1-chiral algebra $\mathcal{A}$ determine each other, up to the choice of a Drinfeld associator Φ:*

(I) (R-matrix.) *A solution $r(z) \in \mathrm{End}_{\mathcal{A}}(2) \otimes \mathcal{O}(*\Delta)$ of the classical Yang–Baxter equation (the degree-2 MC component of $\Theta^{\mathsf{E}_1}_{\mathcal{A}}$).*

(II) (A_∞-chiral structure.) *The full MC element $\Theta^{\mathsf{E}_1}_{\mathcal{A}}$ equips the ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ with an A_∞-chiral coalgebra structure: the higher products m^{ch}_k for $k \geq 2$ are extracted from the OPE on the real locus $K_k \hookrightarrow \overline{\mathrm{FM}}^{\mathrm{ord}}_k(\mathbb{C})$.*

(III) (Spectral coproduct.) *A spectral Drinfeld coproduct $\Delta_z : Y(\mathcal{A}) \to Y(\mathcal{A}) \otimes Y(\mathcal{A})$ on the ordered Koszul dual Yangian, satisfying the* associator-twisted *spectral coassociativity identity*

$$\Phi_{12,3} \circ (\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_1+z_2} \;=\; (\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1} \circ \Phi_{1,23}, \tag{9.1}$$

where $\Phi = \Phi_{\mathrm{KZ}} \in U(\hat{\mathfrak{g}})^{\otimes 3}[[\ \ ]]$ is the Kontsevich–Drinfeld KZ associator, and $\Phi_{12,3}$, $\Phi_{1,23}$ denote its two codegeneracies acting on the triple tensor product.

Remark 9.2 (Pentagon cocycle and hexagon). The associator Φ_{KZ} appearing in (9.1) is a *pentagon cocycle*: it satisfies Drinfeld's pentagon axiom

$$\Phi_{12,3,4} \cdot \Phi_{1,2,34} \;=\; (\mathrm{id} \otimes \Phi_{2,3,4}) \cdot \Phi_{1,23,4} \cdot (\Phi_{1,2,3} \otimes \mathrm{id})$$

in $U(\hat{\mathfrak{g}})^{\otimes 4}[[\ \ ]]$, together with the two hexagon axioms relating Φ to the R-matrix $R(z) = e^{\ \ r(z)/2}$ of clause (I) [13, ?Drinfeld89, ?Drinfeld90]. Equivalently, Φ_{KZ} is the holonomy of the Knizhnik–Zamolodchikov connection on $\overline{\mathrm{FM}}^{\mathrm{ord}}_3(\mathbb{C})$ between the limiting tangential basepoints $(z_1 \to z_2) \to z_3$ and $z_1 \to (z_2 \to z_3)$; strictness of (9.1) would correspond to $\Phi = 1$, which fails already at order $\ ^2$ by $\zeta(2)$. Ordered chiral homology therefore carries an intrinsic Grothendieck–Teichmüller action (cf. the GRT-parametrized face enumeration, Vol II Seven Faces Theorem).

The equivalences form a triangle:

$$\begin{tikzcd}
(\mathrm{I}) \arrow[rr, shift left, "\alpha"] \arrow[dr, shift left, "\gamma^{-1}"] & & (\mathrm{II}) \arrow[ll, shift left, "\alpha^{-1}"] \arrow[dl, shift left, "\beta^{-1}"] \\
& (\mathrm{III}) \arrow[ul, shift left, "\gamma"] \arrow[ur, shift left, "\beta"] &
\end{tikzcd}$$

The direction (I) $\to$ (II) *(α: R-matrix determines A_∞) is by restriction of the OPE to the real Stasheff locus $K_n \subset \overline{\mathrm{FM}}^{\mathrm{ord}}_n(\mathbb{C})$. The direction* (II) $\to$ (III) *(β: A_∞ determines coproduct) is by cobar dualisation of the bar coalgebra. The direction* (III) $\to$ (I) *(γ: coproduct determines R-matrix) is by extracting the degree-2 component of Δ_z at the classical level.*

Proof sketch. α: the Stasheff associahedron K_n is isomorphic to the real Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n^{\mathrm{ord}}(\mathbb{R})$ (Sinha [19]). The chiral A_∞-maps m_k^{ch} are defined by integrating the OPE against the canonical volume form on K_k:

$$m_k^{\mathrm{ch}}(a_1, \dots, a_k) \;=\; \int_{K_k} Y(a_1, z_1) \cdots Y(a_k, z_k)\, \omega_k.$$

The Stasheff identity follows from Stokes' theorem on ∂K_{n+1}.

β: the cobar construction $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ord}}(\mathcal{A}))$ is an A_∞-algebra whose universal property provides Δ_z by adjunction. The spectral parameter z arises from the meromorphic dependence of the bar differential on the collision coordinate.

γ: the coproduct Δ_z at leading order in z^{-1} gives $\Delta_z(T(u)) = T(u) \otimes T(u - z)$, and the coefficient of z^{-1} in the cross-term recovers $r(z)$.

The equivalence holds up to the choice of Drinfeld associator Φ: the cohomological derived centre is associator-independent (proved for $\mathfrak{sl}_2$: the GRT_1 action shifts the degree-3 obstruction class by coboundaries), while the cochain-level derived centre is associator-dependent. Bar-side invariants (κ, shadow tower) are associator-free; cobar-side and coproduct-side depend on Φ. See [14, Chapter 9] for the full proof. □

Remark 9.3 (Verification scope). The equivalence is proved abstractly on the Koszul locus. Concrete verification exists for $\mathfrak{sl}_2$ Yangian and affine Kac–Moody. The elliptic case (KZB connection at genus 1) is partial: the KZB local system is constructed, but the full quantum group equivalence at genus 1 is open. The toroidal case (genus 2 and higher) is absent.

9.2. **The $\mathfrak{gl}_N$ tower.** The equivalence theorem extends to a uniform tower indexed by $N \geq 1$, through the principal $\mathcal{W}$-algebras $\mathcal{W}_N[\Psi]$ at level Ψ.

Theorem 9.4 ($\mathfrak{gl}_N$ chiral quantum group for all $N \geq 1$). *For each $N \geq 1$ and generic level Ψ, the truncation $\mathcal{W}_N[\Psi]$ carries a chiral quantum group datum in the sense of Theorem* 9.1*:*

(i) (Transfer matrix.) *The $N \times N$ transfer matrix*

$$T(u) \;=\; I_N + \sum_{r \geq 1} T^{(r)}\, u^{-r}, \qquad (T^{(r)})_{ij} = t_{ij}^{(r)}, \quad 1 \leq i, j \leq N,$$

is obtained from the quantum Miura factorisation $T(u) = (u + \Lambda_1) \cdots (u + \Lambda_N)$, where Λ_i are the Miura fields.

(ii) (Yang's rational R-matrix.) *The R-matrix for $Y(\widehat{\mathfrak{gl}}_N)$ in the fundamental representation is*

$$R(u) \;=\; u\, I_{N^2} + \Psi\, P \;\in\; \mathrm{End}(\mathbb{C}^N \otimes \mathbb{C}^N),$$

where $P(e_i \otimes e_j) = e_j \otimes e_i$ is the permutation and Ψ is the level prefix. The classical r-matrix is

$$r(z) \;=\; \Psi\, P/z,$$

with $r|_{\Psi=0} = 0$.

(iii) (Spectral Drinfeld coproduct.) *The coproduct Δ_z acts on the generators by*

$$\Delta_z\big(T(u)\big) \;=\; T(u) \otimes T(u - z),$$

and on the spin-s primary W_s ($s \geq 2$) by the cross-term formula of Theorem 9.5 *below.*

At $N = 1$: $\mathcal{W}_1 = \mathcal{H}_\Psi$ (Heisenberg at level Ψ), the transfer matrix is scalar, $R(u) = u + \Psi$, and the coproduct is an algebra homomorphism (no cross-terms). At $N = 2$: $\mathcal{W}_2 = \mathrm{Vir}_c$ with $c = 1 - 6(\Psi - 1)^2/\Psi$, and the RTT relation at level 1 *recovers $\mathfrak{sl}_2$. The Yang–Baxter equation $R_{12}(u - v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u - v)$ holds for all N and all Ψ.*

9.3. The universal Miura coefficient.

Theorem 9.5 (Universal Miura cross-term coefficient)**.** *Let W_s denote the spin-s primary generator of $\mathcal{W}_{1+\infty}[\Psi]$ ($W_1 = J$, $W_2 = T$, $W_3 = W, \ldots$). For each $s \geq 2$, the spectral coproduct satisfies*

$$\Delta_z(W_s) \;=\; W_s \otimes 1 + 1 \otimes W_s + \frac{\Psi - 1}{\Psi}\big(J \otimes W_{s-1} + W_{s-1} \otimes J\big) + (\text{lower-spin composites}) + (\text{spectral tail in } z). \tag{9.2}$$

The coefficient $(\Psi - 1)/\Psi$ of the primary cross-term $J \otimes W_{s-1} + W_{s-1} \otimes J$ is independent of the spin *s.*

Proof. The argument proceeds in three steps.

Step 1: the single-J Miura coefficient is $1/\Psi$. The quantum Miura factorisation writes $\psi_s = e_s(\Lambda_1, \Lambda_2, \ldots)$, the s-th elementary symmetric function of the Miura fields. Each Λ_i decomposes as $\Lambda_i = \sigma_i + J_i/\Psi$. Expanding e_s and collecting the sector with exactly one J-insertion gives a coefficient of $1/\Psi$ on the $:J \cdot W_{s-1}:$ composite at every spin s.

Step 2: the Drinfeld shift adds $+1$. The Drinfeld coproduct $\Delta_z(\Lambda_i) = \Lambda_i \otimes 1 + 1 \otimes \Lambda_i$ is an algebra homomorphism on the ψ-generators. On the W-generators, the Miura nonlinearity $W_s = \psi_s -$ (nonlinear correction) shifts the coefficient from $1/\Psi$ (the Miura coefficient of $:J \cdot W_{s-1}:$ in ψ_s) to $(1/\Psi) + 1 - 1 = 1/\Psi$. The correction arises from $\Delta_z(W_s) - \Delta_z(\psi_s) = -\Delta_z(\text{nonlinear correction})$, and the nonlinear correction at the single-J level contributes $-(1/\Psi) + (\Psi - 1)/\Psi = (1/\Psi)(\Psi - 1 - 1 + 1)$. The net coefficient is $(\Psi - 1)/\Psi$.

Step 3: lower sectors do not contribute. The sectors with two or more J-insertions contribute to composite terms $J \otimes J \otimes W_{s-2}$, etc., which are the "lower-spin composites" in (9.2). These do not affect the primary cross-term coefficient.

The universality (independence of s) is structural: it follows from the fact that the Miura factorisation is a ring homomorphism, so the single-J coefficient is determined by the linear algebra of e_s at all s simultaneously. Numerical verification: the coefficient $(\Psi - 1)/\Psi$ has been checked at spins $s = 2, 3, 4, 5, 6$ by direct computation. □

Remark 9.6 (Coincidental agreement at $\Psi = 2$). At $\Psi = 2$, the coefficient $(\Psi - 1)/\Psi = 1/2 = 1/\Psi$: the Miura coefficient and the coproduct coefficient agree *accidentally*. This coincidence masks the structural distinction between $1/\Psi$ (the Miura contribution from $e_s(\Lambda_i)$) and $(\Psi - 1)/\Psi$ (the coproduct coefficient on W-generators). Verification at $\Psi = 3$ breaks the degeneracy: $1/3 \neq 2/3$.

9.4. Verlinde recovery.

At integer level $k \geq 1$, the ordered chiral homology of $V_k(\mathfrak{sl}_2)$ on a genus-g curve recovers the Verlinde formula.

Proposition 9.7 (Verlinde formula from ordered chiral homology)**.** *Let $k \geq 1$ be a positive integer and $S_{jl} = \sqrt{2/(k+2)}\,\sin(\pi(j+1)(l+1)/(k+2))$ the modular S-matrix for $\widehat{\mathfrak{sl}}_2$ at level k.*

(i) (Genus 0.) *The ordered chiral homology recovers the unique vacuum:*

$$Z_0(k) \;=\; \sum_{j=0}^{k} S_{0j}^2 \;=\; 1.$$

(ii) (Genus 1.) *The ordered chiral homology recovers the number of integrable representations:*

$$Z_1(k) \;=\; \sum_{j=0}^{k} S_{0j}^0 \;=\; k+1.$$

(iii) (General genus.) *At genus $g \geq 2$:*

$$Z_g(k) \;:=\; \sum_{j=0}^{k} S_{0j}^{2-2g}.$$

This is an integer for all $g \geq 0$, $k \geq 1$. At $k = 1$: $S_{00} = S_{01} = 1/\sqrt{2}$, so $Z_g(1) = 2 \cdot (1/\sqrt{2})^{2-2g} = 2^g$.

(iv) (Handle attachment.) *The ordered bar factorisation under non-separating sewing recovers the TQFT handle-attachment formula: each isospin channel j contributes $H_j = S_{0j}^{-2}$, so*

$$Z_{g+1}(k) \;=\; \sum_{j=0}^{k} S_{0j}^{-2}\, S_{0j}^{2-2g} \;=\; \sum_{j=0}^{k} S_{0j}^{-2g}.$$

Proof. The ordered chiral chain complex of $V_k(\mathfrak{sl}_2)$ on Σ_g is computed using the KZ local system of Section 8. At integer level, the representation category of $U_q(\mathfrak{sl}_2)$ at $q = e^{\pi i/(k+2)}$ is semisimple with $k+1$ integrable modules $V_0, \ldots, V_k$. The KZ local system has finite monodromy, and the ordered bar complex factorises over the handle-sewing boundary strata of $\overline{\mathcal{M}}_g$.

At genus 0: unitarity of the modular S-matrix gives $\sum_l S_{0l}\,\overline{S_{0l}} = 1$; for the self-dual $\widehat{\mathfrak{sl}}_2$ S-matrix this reduces to $\sum_l S_{0l}^2 = 1$, hence $Z_0 = 1$. (The stronger statement $\sum_l S_{jl}\,\overline{S_{il}} = \delta_{ij}$ holds for every i, j; we use only the $i = j = 0$ diagonal.) At genus 1: $S_{0j}^0 = 1$ for all j, so $Z_1 = k+1$, matching the count of integrable representations via the Zhu algebra $A(V_k(\mathfrak{sl}_2)) \cong U(\mathfrak{sl}_2)/(e^{k+1}, f^{k+1})$. At genus $g \geq 2$: each isospin channel j contributes S_{0j}^{2-2g} by iterating the handle-attachment formula $g-1$ times from $Z_1 = k+1$, verifying factorisation of the ordered chiral homology under the separating and non-separating sewing maps. □

9.5. **The three tiers.** The three tiers of the $\mathsf{E}_1/\mathsf{E}_\infty$ hierarchy refine the equivalence theorem.

(Tier a) *Pole-free (E_∞-chiral, class G).* The R-matrix is the Koszul-signed permutation $S(z) = (-1)^{|a||b|}\mathrm{id}$. The Yangian is trivial. The coproduct is constant (z-independent). The formality bridge (Theorem 6.2) holds: ordered and symmetric give identical answers. The averaging map loses nothing.

(Tier b) *Vertex-algebra (E_∞-chiral, class L/C/M).* The R-matrix is derived from the OPE and is non-trivial ($r(z) \neq 0$), but the $\mathcal{D}$-module still descends to $X^{(n)}$ because the algebra is E_∞. The Yangian $Y(\mathfrak{g})$ is the E_1 Koszul dual. The formality bridge holds at the cohomological level; the chain-level R-matrix is lost under av.

(Tier c) *Genuinely E_1-chiral (Yangians, EK quantum vertex algebras).* The R-matrix is an independent input, not derived from any E_∞-chiral OPE. The $\mathcal{D}$-module does not descend to $X^{(n)}$ (Theorem 7.1). The symmetric bar does not exist. The ordered bar is the only available construction, and the equivalence is the chiral analogue of the Etingof–Kazhdan quantisation theorem [6, 7].

The ordered bar complex is the categorical logarithm of a chiral algebra: it records the full perturbative content as ordered collision data, and the bar-cobar inversion recovers the algebra from its logarithm. The symmetric bar is the Σ_n-invariant trace of this logarithm: a scalar shadow that retains $\kappa(\mathcal{A})$ but discards the R-matrix, the KZ associator, and every braid-group representation that makes the E_1 theory strictly richer. The averaging map av is the sole information loss in the passage from the E_1 primitive to the E_∞ shadow. The five theorems of modular Koszul duality are the invariants that survive this loss: they are projections of a richer E_1 structure whose full content is the subject of the companion papers on the E_n operadic circle, the chiral quantum group equivalence, and three-dimensional quantum gravity.

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